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Artificial Intelligence

Machine Learning Models for Predicting Treatment-Requiring Retinopathy of Prematurity in the e-ROP Study

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Purpose: To evaluate machine learning (ML) models for predicting treatment-requiring retinopathy of prematurity (TR-ROP) using image findings at 32 to 34 weeks of postmenstrual age, along with demographic and clinical characteristics.

Methods: This secondary analysis included 771 infants with a birth weight of less than 1251 g who had at least one imaging session by 34 weeks postmenstrual age and at least one subsequent ROP examination for determining TR-ROP by ophthalmologists in the Telemedicine Approaches to Evaluating Acute-Phase Retinopathy of Prematurity (e-ROP) Study. Six ML models (K-nearest neighbors, support vector machine, random forest, extreme gradient boosting, deep neural network [DNN], and transformer) were evaluated for predicting TR-ROP. Prediction performance was evaluated using the area under the receiver operating characteristic curve (AUC), accuracy, sensitivity, and specificity.

Results: Using image findings and demographic and clinical data, ML models achieved AUCs ranging from 0.777 (K-nearest neighbors) to 0.853 (DNN), sensitivity ranging from 0.765 (extreme gradient boosting) to 0.929 (DNN), and specificity ranging from 0.644 (DNN) to 0.698 (transformer). Using image findings alone, the DNN performed best with an AUC of 0.787, sensitivity of 0.729, and specificity of 0.725.

Conclusions: ML models using image findings, demographics and clinical characteristics moderately predict TR-ROP, with DNN achieving the highest AUC and sensitivity. Although ML models may provide tools for the early identification of high-risk infants for close monitoring and timely treatment of TR-ROP, future research is needed to improve their performance.

Translational Relevance: ML has the potential to predict TR-ROP risk based on early image findings, demographics, and clinical characteristics.

Introduction

Retinopathy of prematurity (ROP), a serious vasoproliferative disease of the retina of premature infants, is the leading cause of a treatable disease that may lead to blindness in children worldwide. Treatment-requiring ROP (TR-ROP) can usually be effective when treated with retinal ablative surgery or anti-vascular endothelial growth factor if diagnosed

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timely within the acute phase time window.[1] Monitoring the development of TR-ROP usually involves subjecting at-risk infants to medical- and resource-intensive, and thereby costly, serial diagnostic eye examinations performed by ophthalmologists, beginning at 32 to 34 weeks postmenstrual age (PMA) regardless of gestational age at birth. In the United States, ROP screening guidelines require examining all infants with a birth weight (BW) of 1500 grams or less or a gestational age of 30 weeks or less, as well as a select number of infants outside of these criteria as determined by the caregivers.[2] However, this guideline results in a number of unnecessary examinations because fewer than 10% of infants meet the ROP treatment criteria.[1,3] Telemedicine systems have been found to be safe and effective for ROP screening.[4–8] Our e-ROP (Telemedicine Approaches to Evaluating Acute-Phase Retinopathy of Prematurity) study demonstrated that a telemedicine system using nonphysician imagers and nonphysician trained readers for detection of referral-warranted ROP (RW-ROP, defined as zone I disease, stage 3 disease or worse, or plus disease) is valid, safe, and cost effective.[6,9–11]

The image evaluation of ROP usually involves determining the presence of ROP morphological features, including preplus or plus disease, location of ROP (zones I, II, or III), and ROP stage. Our previous investigations using traditional regression models showed that ROP morphological findings of an eye at the first image session performed by the nonphysician by 34 weeks PMA predicted the subsequent development of TR-ROP[12] beyond what is already predicted by the BW, gestational age, postnatal weight gain, and respiratory support status at the examination. However, the area under ROC curve (AUC) from such a prediction model was not excellent (0.88 for predicting TR-ROP) with a sensitivity of 98.8% and specificity of 40.1% using the risk score cutoff of 3 points. In light of this, there is a clear need to evaluate whether machine learning (ML) models can improve ROP prediction. ML models have great potential to integrate complex data patterns more effectively than traditional regression methods, thereby improving the accuracy of ROP predictions. This advancement could lead to more targeted and efficient ROP screening, decreasing the burden of unnecessary ROP examination while optimizing ROP care for at-risk infants.

The purpose of this study was to evaluate various ML models for predicting TR-ROP using image findings at 32 at 34 PMA weeks along with data on demographic and clinical characteristics by performing the analysis of unique data from the multicenter e-ROP study.[6,13]

Methods

This is a secondary analysis of the data from the e-ROP Study, a multicenter observational cohort study to evaluate the validity, reliability, feasibility, safety and relative cost effectiveness of a digital imaging system to identify infants with RW-ROP. The details of the design, methods, and primary results for the e-ROP Study have been previously published.[6,9–11] Major features of the image evaluation procedure and the predictors for TR-ROP relevant to this study are described here.

Study Participants

Infants eligible for enrollment into the e-ROP Study had a BW of less than 1251g, were likely to survive 28 days and to remain in neonatal intensive care units at 13 North American centers for serial ROP examinations and concurrent imaging. Informed consent was obtained from the parent or guardian of eligible infants. Institutional review boards approved the study protocol and informed consent process of participating centers.

Study Procedure

Infants underwent serial ROP imaging in both eyes using the RetCam Shuttle (Clarity Medical Systems, Pleasanton, CA), in addition to a standard diagnostic examination by Study-certified ophthalmologists experienced in diagnosing ROP.[6] The ROP findings (stage, zone, and preplus/plus disease) from eye examinations were summarized using the 2005 International Classification of ROP.[14] From each eye, a trained imager obtained an image set that included an external image to assess pupillary dilation, along with five retinal views: disc center and four disc off-centered giving views of the inferior, superior, temporal, and nasal retina. The imagers were masked to the results of the examination and the ophthalmologists were masked to the images. Timing of first ROP examinations and concurrent imaging was determined by local Clinical Center criteria for usual clinical care and imaging sessions began at 32 weeks PMA.

Follow-up diagnostic examinations and concurrent imaging were conducted at least every other week unless medically contraindicated until the ophthalmologist noted one of the following: mature retinal vessels, immature zone III on two occasions at least 7 days apart, ROP regressed or regressing on two visits at least 7 days apart, treatment for severe ROP, or the infant reached 40 weeks PMA with no ROP or only stage 1

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or 2 ROP.[6,13] For infants who were discharged to other nonstudy clinical centers before meeting any of these study end points, follow-up information was obtained to determine whether the infant was treated for ROP.

Image Evaluation

All image sets were graded by nonphysician trained readers in a centralized image reading center at the University of Pennsylvania following the standard grading protocol.[9] Each image set was graded independently by two trained readers with discrepancies adjudicated by the reading supervisor. Readers were masked to the results of diagnostic examinations, previous gradings for either eye of the infant, images of the fellow eye, and demographic data.[9]

All readers determined ROP based on the presence of a demarcation line (stage 1), a ridge (stage 2), extraretinal neovascularization (stage 3), or retinal detachment (stage 4). In addition, the reader-determined zone of vascularization or the zone in which morphological features consistent with ROP were present, and, for each quadrant, whether the posterior pole vessels were normal or sufficiently abnormal to be designated as preplus or plus disease. In a secondary analysis of e-ROP images, all image sets ($n = 1705$) from eyes ever noted to have hemorrhage either on examination by a study-certified ophthalmologist or on initial e-ROP image grading were selected for detailed hemorrhage grading for the presence, location, and type of retinal hemorrhages (dot, blot, flame, or preretinal).[15] When the image was not of sufficient quality to grade specific ROP features, it was recorded as cannot grade.

Data Preprocessing for ML Models

In distance-based ML algorithms like k-Nearest Neighbors (KNN) and support vector machines (SVM), the scale of feature dimensions significantly affects model performance. In the e-ROP study dataset, BWs ranged from 380 to 1250g, and gestational age ranged from 22 to 32 weeks. These disparate scales can disproportionately impact distance computations, introducing scale-induced bias into the ML algorithm. To address this issue, normalization techniques are often used. In this study, we apply z-score normalization, mathematically defined as:

$$z = \frac{x - \mu}{\sigma},$$

where z is the normalized score, x represents the original feature value, μ is the mean of the feature, and σ is the standard deviation. By standardizing the features, we ensure that no individual feature unfairly influences distance calculations. This strategy improves the robustness and generalizability of the ML performance.

ML Methods

To rigorously investigate the ML models for predicting RW-ROP, a diverse set of ML algorithms was used. These algorithms included both traditional ML methods and modern deep learning techniques, each with their own customizations and hyperparameters that were subject to tuning. The selection of these algorithms was motivated by their proven effectiveness in analyzing healthcare data characteristics. It is worth mentioning that the deep neural network (DNN) and transformer models were custom developed for this study using PyTorch, while the foundational frameworks for these algorithms were open source. Specifically, the ML models used were as follows:

(1) K-nearest neighbors (KNN): The KNN algorithm is an instance-based learning method. It scans the entire dataset to identify the K instances that are most similar to a given query point, typically based on a distance metric. The algorithm's output is then determined by a majority vote of its K neighbors. Despite its computational complexity, KNN has demonstrated success in a range of applications, including classification, regression, and recommendation systems.

(2) SVM-radial basis function (RBF): SVMs are a type of supervised learning model frequently used for classification and regression tasks. These models locate a hyperplane that optimally separates the data into distinct classes. Although linearly separable data can be accurately classified by standard SVMs, incorporating an RBF kernel can enhance their efficacy. The incorporation of the RBF kernel enables the SVM to operate in a transformed, higher-dimensional feature space, facilitating the separation of data that is not linearly separable in the original feature space. This expansion has demonstrated its versatility in various types of classification problems.

(3) Random Forest: Random forest is an ensemble learning method that builds on the concept of decision trees. A decision tree is a model resembling a flowchart, used for classification or regression tasks, and makes decisions based on feature values. Random Forest constructs multiple decision trees during the training phase and aggregates its predictions for the final output.

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Specifically, the final prediction is either an average or a majority vote of the individual trees' predictions. Each tree in the ensemble is trained on a bootstrap sample drawn with replacement from the original dataset, and a random subset of features is selected for each split in the tree. This stochastic method improves the model's robustness and generalizability, enhancing its effectiveness in a wide range of applications.

(4) Extreme gradient boosting (XGBoost): XGBoost is an optimized version of the gradient boosting algorithm, designed for both computational efficiency and high performance. It builds on the standard gradient boosting framework by incorporating regularization terms to prevent overfitting. XGBoost has been widely acclaimed for its performance in Kaggle competitions and is especially effective for structured data and imbalanced datasets.

(5) DNNs: DNNs are a class of artificial neural networks that use multiple hidden layers to process input data in a sequential manner, enabling them to model complex patterns in the data. In this study, the DNN was designed for binary classification, consisting of an input layer, multiple hidden layers, and an output layer. The input layer projected the input data to a hidden representation. The hidden layers, equipped with ReLU activation functions, captured hierarchical feature representations through linear and nonlinear transformations. Dropout regularization was applied after each hidden layer to mitigate overfitting. The output layer contained two neurons with a log-softmax activation function to compute class probabilities. To optimize model performance, hyperparameters were systematically tuned using a grid search strategy. Specifically, the number of hidden layers varied from 1 to 3, while the number of neurons per hidden layer was selected from {4, 8, 16, 32, 64, 128}. The dropout rate was tuned within the range [0, 0.4] in increments of 0.1. The training was conducted using the Adam optimizer, with the learning rate chosen from {0.001, 0.005, 0.01} and weight decay from $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$. Negative log likelihood was used as the loss function, and early stopping with a patience of 50 epochs was applied based on validation AUC to prevent overfitting. Mini-batch training was used with batch sizes selected from {16, 32, 64}.

(6) Transformer: Transformers are a class of deep learning models that use self-attention mechanisms to effectively capture long-range dependencies and complex feature interactions in the data. In this study, a custom transformer architecture was designed for binary classification. The architecture consisted of four main components: an input embedding layer, a positional encoding module, a transformer encoder, and a classification head. The input embedding layer projected input features into a high-dimensional space, and the positional encoding module incorporated sequence order information to preserve positional relationships between features. The transformer encoder, consisting of stacked multihead self-attention layers and feed-forward networks, captured model dependencies and interactions among features. Finally, the classification head aggregated the encoder's outputs using mean pooling and mapped the resulting representation to two logits with a log-softmax activation function for binary classification. To optimize performance, hyperparameters were systematically tuned using a grid search strategy. The hidden dimensions were selected from {8, 16, 32}, whereas the number of encoder layers and attention heads were chosen from {2, 4, 8, 16}. These combinations were evaluated across different feature sets to determine the optimal configuration for prediction. The model was trained using the Adam optimizer, with the learning rate selected from {0.001, 0.005, 0.01} and weight decay from $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$. Negative log likelihood was used as the loss function. Early stopping with a patience of 50 epochs was applied to prevent overfitting, using validation AUC as the monitoring metric. Minibatch training was performed with batch sizes chosen from {16, 32, 64}.

Each ML model has specific hyperparameters and underlying assumptions that have been carefully tuned and validated. To ensure robust performance, the hyperparameters were tuned through iterative validation using different data splits.

ML Model Evaluation

To conduct an objective and precise evaluation of the ML models for predicting TR-ROP, we used a detailed 10-fold cross-validation strategy. Initially, the dataset was divided into 10 nonoverlapping subsets through a stratified shuffle split, which preserves the original class distribution within each subset. In each fold, one subset was designated as the test set and another as the validation set, while the remaining

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eight subsets were used for training, ensuring an 8:1:1 ratio for training, validation, and testing. This setup facilitates both hyperparameter optimization on the validation set and testing on the test set. Each model was fine-tuned independently in each fold using a comprehensive search over a predefined hyperparameter space, and its performance was assessed based on the validation outcomes for that fold. The performance metrics from the 10 test sets were then combined and used to calculate performance metrics, providing an unbiased evaluation of the model's effectiveness.

The models' performance was assessed using four metrics: the AUC, the area under the precision-recall curve (AUPRC), accuracy, sensitivity, and specificity. These metrics were computed for each test set and then averaged over 10 iterations to yield dependable performance estimates. This approach minimizes the risk of overfitting and improves the generalizability of the performance metrics. It adds analytical rigor to our methodology, especially for smaller datasets and models that demand meticulous hyperparameter tuning.

Feature Selection

For each ML algorithm used in this study, we executed the following steps repeatedly:

(1) Feature sets: Five sets of features were considered as predictors for ML models: (1) only includes image evaluation findings (preplus/plus disease, number of quadrants with preplus, stage of ROP, zone of ROP, blot hemorrhage, preretinal hemorrhage); (2) only includes demographics (BW, gestational age, gender, race, and ethnicity); (3) only includes clinical characteristics (respiratory status, feeding status, and weight gain); (4) using both demographics and clinical characteristics; and (5) includes image evaluation findings, demographics, and clinical characteristics.
(2) Model training: Each ML model was trained for predicting the development of TR-ROP in infants, using five feature sets.
(3) Model validation: The performance of each ML model was validated using the previously described two-step data partitioning approach (refer to the "ML Model Evaluation" section).

By following this procedure, we conducted a thorough evaluation of the feature sets and ML models, thereby strengthening the durability and applicability of our results.

Results

Characteristics of Study Infants

Among 1284 infants enrolled in the e-ROP study, 771 (60%) had an imaging session by 34 weeks PMA and at least one subsequent ROP examination for determining TR-ROP by study-certified ophthalmologists, and thus were included in this study. During the follow-up, 85 infants (11.0%) developed TR-ROP in the judgment of the examining ophthalmologist.

The comparisons of demographic and clinical characteristics between infants with vs. without the development of TR-ROP were reported in Table 1. Compared with infants without the development of TR-ROP, infants with the development of TR-ROP had significantly lower mean BW (679 g vs. 874 g; $P < 0.0001$), lower gestational age (25.0 weeks vs. 26.9 weeks; $P < 0.0001$), and lower postnatal weight gain rate (13.2 g/d vs. 14.6 g/d; $P = .009$), more likely to be Hispanic or Latino (20.0% vs.10.5%; $P = 0.03$), more likely to use controlled mechanical ventilator (35.3% vs. 11.1%; $P < 0.0001$), and less likely to be on full feed (58.8% vs. 73.2%; $P = .0007$). However, there were no significant differences between the two groups of infants in gender ($P = 0.19$), race ($P = 0.43$), or multiple births ($P = 0.84$).

The comparisons of the image evaluation findings from the first image session by 34 weeks PMA between infants with vs. without TR-ROP are shown in Table 2. Compared with infants without TR-ROP, infants with TR-ROP were more likely to have preplus disease (47.1% vs. 16.6%; $P < 0.0001$), a greater number of quadrants with preplus (preplus in all four quadrants: 18.8% vs. 2.5%; $P < 0.0001$), higher stage of ROP (stage 3 or above: 41.2% vs. 10.2%; $P < 0.0001$), lower ROP zone (zone 1: 34.1% vs. 4.7%; $P < 0.0001$), presence of blot hemorrhage (21.2% vs. 4.4%; $P < 0.0001$), and preretinal hemorrhage (11.8% vs. 2.9%; $P < 0.0001$).

Performance of ML Models for Predicting TR-ROP Using Demographics, Clinical Characteristics, and Image Evaluation Findings

The performances of ML models in terms of AUC, AUPRC, accuracy, sensitivity, and specificity at various cutoffs were provided in Table 3 and their receiver operating characteristic curves were shown in Figure 1. The models had AUC ranging from 0.777 (KNN) to 0.853 (DNN), and AUPRC ranges from 0.314 (KNN) to 0.397 (DNN). The random forest model performed

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Table 1. The Demographic and Clinical Characteristics of Infants With vs. Without Developing TR-ROP ($n = 771$)

Infant Characteristics	Without TR-ROP ($n = 686$)	With TR-ROP ($n = 85$)	*P* Value
BW (g): Mean (SD)	874 (203)	679 (165)	<0.0001
Gestational age (wk): Mean (SD)	26.9 (1.8)	25.0 (1.5)	<0.0001
PMA at the first image session (wk): Mean (SD)	33.4 (0.9)	33.1 (0.9)	0.003
Weight growth rate (g/d) by the first image session: Mean (SD)	14.6 (4.3)	13.2 (4.7)	0.009
Gender: female (%)	334 (48.7%)	35 (41.2%)	0.19
Ethnicity of infant: n (%)			0.03
Not Hispanic or Latino	581 (84.7%)	66 (77.7%)	
Hispanic or Latino	72 (10.5%)	17 (20.0%)	
Unable to answer	33 (4.8%)	2 (2.4%)	
Race of infant: n (%)			0.43
Black only	216 (31.5%)	21 (24.7%)	
White only	379 (55.3%)	51 (60.0%)	
Others	91 (13.3%)	13 (15.3%)	
Multiple births: Yes (%)	209 (30.5%)	25 (29.4%)	0.84
Respiratory support			<0.0001
No respiratory support	166 (14.2%)	7 (8.2%)	
Nasal canula	198 (28.9%)	14 (16.5%)	
CPAP/NIPPV/HFNC	240 (35.0%)	29 (34.1%)	
Controlled mechanical ventilator	86 (11.1%)	30 (35.3%)	
High-frequency oscillatory ventilation	6 (0.9%)	5 (5.9%)	
Feeding status at the first image session			0.0007
Full feeds	502 (73.2%)	50 (58.8%)	
Nil per OS	49 (7.1%)	17 (20.0%)	
Trophic	20 (2.9%)	2 (2.4%)	
Advanced feeds	115 (16.8%)	16 (18.8%)	

SD, standard deviation.

similarly well as DNN with AUC of 0.848 and AUPRC of 0.394.

At the default cutoff of 0.5, all ML models had very high specificity ranging from 0.983 (transformer) to 0.993 (random Forest), yet very poor sensitivity (lower than 0.15). When the optimal cutoff was determined based on the largest sum of sensitivity and specificity, the DNN model achieved high sensitivity (0.929) yet lower specificity (0.644). The random forest had sensitivity of 0.859 and specificity of 0.682. The other models had lower sensitivity ranging from 0.765 (XGBoost) to 0.835 (SVM), and specificity ranging from 0.659 (SVM) to 0.698 (transformer).

Performance of ML Models for Predicting TR-ROP Using Image Evaluation Findings and Other Predictors

When only image evaluation findings at 34 weeks PMA or earlier were used to predict TR-ROP, all performance of ML models became worse. The AUC ranged from 0.609 (SVM) to 0.787 (DNN) (Table 4, Fig. 2), and AUPRC ranges from 0.204 (SVM) to 0.366 (XGBoost). Using the default 0.50 as the cutoff, all ML models achieved high specificity (0.975 or higher), but the sensitivity for each model was very low (0.212 or lower). Using the optimal cutoff specific to each ML model, the sensitivity improved ranging from 0.306 (SVM) to 0.729 (DNN), and specificity ranges from 0.725 (DNN) to 0.939 (SVM).

The ML models that used other predictors did not predict TR-ROP well, with AUC ranging from 0.666 (SVM) to 0.782 (DNN) using demographics only (Table 5), with AUC ranging from 0.528 (SVM) to 0.695 (DNN) using clinical characteristics only (Table 6), and AUC ranging from 0.676 (SVM) to 0.798 (random forest) using both demographics and clinical characteristics (Table 7).

In the best performing model (DNN), the number of quadrants with preplus predicted TR-ROP with an

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Table 2. Findings From the Trained Readers Evaluation of Images Taken at the First Image Session

Image Evaluation Findings in Worse Eye of Specific Finding	Without TR-ROP ($n = 686$)	With TR-ROP ($n = 85$)	*P* Value
Preplus/plus			<0.0001
Normal	564 (82.2%)	37 (43.5%)	
Preplus	114 (16.6%)	40 (47.1%)	
Plus	0 (0.0%)	1 (1.2%)	
Unknown	8 (1.2%)	7 (8.3%)	
No. of quadrants with preplus			<0.0001
0	564 (82.2%)	37 (43.5%)	
1	45 (6.6%)	8 (9.4%)	
2	30 (4.4%)	9 (10.6%)	
3	22 (3.2%)	8 (9.4%)	
4	17 (2.5%)	16 (18.8%)	
Unknown	8 (1.2%)	7 (8.2%)	
Stage of ROP			<0.0001
No	459 (66.9%)	21 (24.7%)	
Stages 1–2	141 (20.6%)	23 (27.1%)	
Stages 3–4	70 (10.2%)	35 (41.2%)	
Unknown	16 (2.3%)	6 (7.1%)	
Zone of ROP			<0.0001
No ROP	459 (66.9%)	21 (24.7%)	
II	179 (26.1%)	29 (34.1%)	
I	32 (4.7%)	29 (34.1%)	
Unknown	16 (2.3%)	6 (7.1%)	
ROP stage and zone			<0.0001
No ROP	459 (66.9%)	21 (24.7%)	
Stages 1–2, Zone II	124 (18.1%)	12 (14.1%)	
Stages 3–4, Zone II	55 (8.0%)	17 (20.0%)	
Stages 1–2, Zone I	17 (2.5%)	11 (12.9%)	
Stages 3–4, Zone I	15 (2.2%)	18 (21.2%)	
Unknown	16 (2.3%)	6 (7.1%)	
Blot hemorrhage			<0.0001
No	656 (95.6%)	67 (78.8%)	
Yes	30 (4.4%)	18 (21.2%)	
Preretinal hemorrhage			<0.0001
No	666 (97.1%)	75 (88.2%)	
Yes	20 (2.9%)	10 (11.8%)	

TR-ROP, treatment-requiring retinopathy of prematurity.

AUC (AUPRC) of 0.720 (0.290), the ROP stage and zone had an AUC of 0.766 (0.333), the combination of the number of quadrants with preplus, ROP stage, and zone had an AUC of 0.769 (0.328), and adding blot hemorrhage and preretinal hemorrhage did not improve the AUC (AUPRC) (0.761 and 0.306 respectively) (Table 8).

The performances of the best performing model, DNN, for predicting TR-ROP using various predictors alone and in combinations are reported in Table 8. The demographic (BW, gestational age), image evaluation findings (number of quadrants with preplus, ROP stage, and zone), and clinical characteristics (respiratory status, feeding status, and weight gain rate) each contributed to the prediction of TR-ROP, with an AUC (AUPRC) of 0.782 (0.332) from demographics, 0.769 (0.328) from image evaluation findings, 0.686 (0.234) from clinical characteristics, and the combination of these predictors provided the better prediction with AUC of 0.828 and AUPRC of 0.372.

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Table 3. The Area Under Receiver Operating Characteristic Curve, Accuracy, Sensitivity, and Specificity From Various ML Models Using Data of Image Evaluation Findings and Demographic and Clinical Characteristics

			Default Cutoff[a] (0.50)			Optimal Cutoff[a] (highest Sen + Sp)			
ML Model	AUPRC	AUC	Accuracy	Se	Sp	Cutoff	Accuracy	Se	Sp
KNN	0.314	0.777	0.888	0.059	0.991	0.07	0.706	0.788	0.695
SVM	0.340	0.812	0.892	0.094	0.991	0.10	0.678	0.835	0.659
Random Forest	0.394	0.848	0.892	0.082	0.993	0.11	0.702	0.859	0.682
XGBoost	0.355	0.798	0.894	0.141	0.987	0.09	0.702	0.765	0.694
DNN	0.397	0.853	0.894	0.129	0.988	0.05	0.676	0.929	0.644
Transformer	0.347	0.806	0.888	0.129	0.983	0.10	0.712	0.824	0.698

AUC, area under the ROC curve; AUPRC, area under the precision recall curve values; Se, sensitivity; Sp, specificity.
[a]Cutoff refers to the predicted probability of TR-ROP from the specific ML model. When the predicted probability is higher than the cutoff, the infant was deemed to develop TR-ROP.

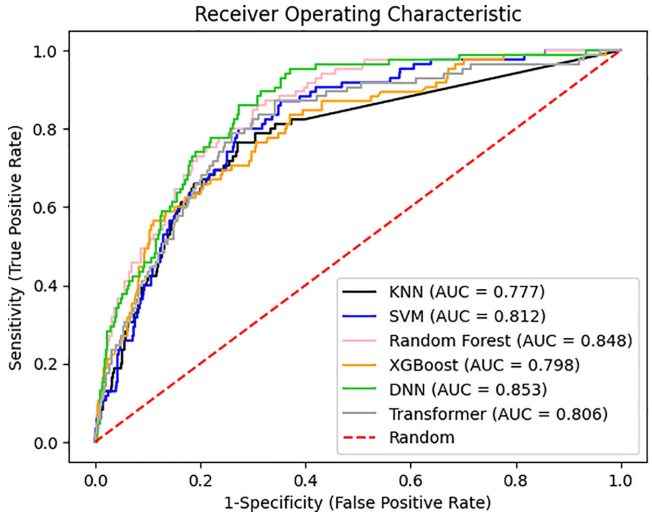


Figure 1. The receiver operating characteristic curves from ML models for predicting TR-ROP image evaluation findings, demographic and clinical characteristics.

Discussion

Using the unique dataset from 771 premature infants in the multicenter e-ROP study, we comprehensively evaluated six commonly used ML models for predicting TR-ROP including KNN, SVM, random Forest, XGBoost, DNN, and transformer, using a variety of predictors (demographics, clinical characteristics, and image findings). We demonstrated that the performance of ML models varied, and the DNN generally performed better than other ML models. However, all the ML models did not achieve high prediction performance, which may not be satisfactory for clinical use. To our understanding, this analysis represents the first attempt to evaluate and compare a variety of ML models for predicting TR-ROP at a single time point (PMA 34 weeks or earlier).

ML models are useful computational tools that enable efficient navigation through complex datasets. However, their performance may not be broadly gener-

Table 4. Sensitivity, Specificity, and Area Under Receiver Operating Characteristic Curve From ML Models Only Using Image Evaluation Findings

			Default Cutoff[a] (0.5)			Optimal Cutoff[a] (Highest Sen + Sp)			
ML Model	AUPRC	AUC	Accuracy	Se	Sp	Cutoff	Accuracy	Se	Sp
KNN	0.219	0.651	0.879	0.106	0.975	0.15	0.794	0.482	0.832
SVM	0.204	0.609	0.888	0	0.999	0.11	0.869	0.306	0.939
Random Forest	0.328	0.782	0.885	0.082	0.984	0.11	0.768	0.647	0.783
XGBoost	0.366	0.778	0.896	0.212	0.981	0.14	0.822	0.600	0.850
DNN	0.348	0.787	0.892	0.129	0.987	0.10	0.725	0.729	0.725
Transformer	0.292	0.768	0.886	0.035	0.991	0.11	0.783	0.671	0.794

[a]Cutoff refers to the predicted probability of TR-ROP from the specific ML model. When the predicted probability is higher than the cutoff, the infant was deemed to develop TR-ROP.

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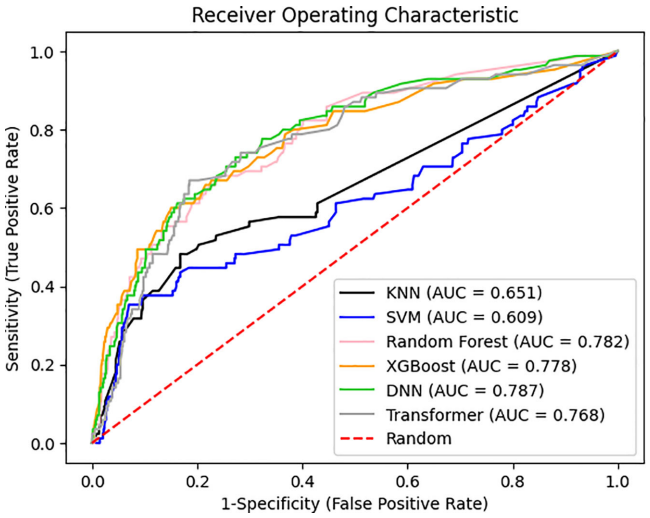


Figure 2. The receiver operating characteristic curves ML models for predicting TR-ROP only using image evaluation findings.

alizable and largely depends on the salient features of the data and the task at hand. Therefore, conducting rigorous experimentation is crucial to identify the best algorithm for specific applications. In this study, we evaluated six ML models that are commonly used. When both demographics (BW, gestational age), clinical characteristics (postnatal weight gain, respiratory support, feeding status), and single-time image evaluation findings for posterior pole and ROP by trained readers were used in ML for prediction of TR-ROP, we found DNN performed best with AUC 0.853, sensitivity of 0.929, and specificity of 0.644, while the KNN performed worst with AUC of 0.777, sensitivity of 0.788, and specificity of 0.695. This single-time earlier prediction by 34 weeks PMA is crucial for identifying high-risk infants that need close monitoring for the development of TR-ROP for timely treatment. Although the DNN model yielded the highest sensitivity of 0.929, the specificity is low (0.644) using the optimal cutoff of the predicted probability of TR-ROP, which may not be satisfactory for clinical use because of the false positives and false negatives can have significant implications in the clinical care of infants. The suboptimal performance of all six ML models indicate that these ML models needs improvement to be used as a tool for identifying high-risk infants for close follow-up for progression to TR-ROP.

In ML prediction, adjusting the threshold of the predicted probability of a disease for optimizing sensitivity and specificity is crucial for clinical application. Lowering the threshold may boost the sensitivity, which ensures the identification of most cases of TR-ROP, but also has the potential to increase the number of false positives. Conversely, a higher threshold may improve specificity, reducing the number of false-positive identifications but increasing the risk of false negatives (i.e., TR-ROP was missed). In this study, we found the default cutoff (0.50) of predicted probability for sensitivity and specificity may not be optimal, the optimal cutoff to achieve the desirable sensitivity (e.g., 100%) may need to be determined for clinical application. The selection of the threshold will rely on the clinical context and the significance of sensitivity versus specificity in the identification of high-risk infants of TR-ROP. Moreover, different ML models may have different optimal cutoffs for achieving a balance between sensitivity and specificity. Consequently, empirical studies are necessary to determine the most suitable thresholds and be validated in an independent dataset before its application in specific clinical settings.

Detection of TR-ROP usually involves subjecting at-risk infants to resource-intensive diagnostic eye examinations performed by ophthalmologists.

Table 5. Sensitivity, Specificity, and Area Under Receiver Operating Characteristic Curve From ML Models Only Using Demographics[§]

			Default Cutoff* (0.5)			Optimal Cutoff* (Highest Sen + Sp)			
ML Model	AUPRC	AUC	Accuracy	Se	Sp	Cutoff	Accuracy	Se	Sp
KNN	0.217	0.677	0.865	0.129	0.956	0.08	0.711	0.600	0.725
SVM	0.232	0.666	0.887	0	0.997	0.12	0.872	0.271	0.946
Random Forest	0.311	0.758	0.881	0.035	0.985	0.15	0.777	0.671	0.790
XGBoost	0.318	0.760	0.888	0.071	0.990	0.16	0.765	0.647	0.780
DNN	0.328	0.782	0.894	0.094	0.993	0.10	0.702	0.729	0.698
Transformer	0.241	0.735	0.890	0	1	0.11	0.759	0.671	0.770

[§]Demographics include BW, gestational age, gender, race, and ethnicity

*Cutoff refers to the predicted probability of TR-ROP from the specific ML model. When the predicted probability is higher than the cutoff, the infant was deemed to develop TR-ROP.

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Table 6. Sensitivity, Specificity, and Area Under Receiver Operating Characteristic Curve From ML Models Only Using Clinical Characteristics[§]

			Default Cutoff[*] (0.5)			Optimal Cutoff[*] (Highest Sen + Sp)			
ML Model	AUPRC	AUC	Accuracy	Se	Sp	Cutoff	Accuracy	Se	Sp
KNN	0.155	0.569	0.860	0.094	0.955	0.09	0.674	0.459	0.701
SVM	0.163	0.528	0.891	0.012	1	0.12	0.883	0.082	0.983
Random Forest	0.206	0.684	0.887	0	0.997	0.10	0.621	0.682	0.614
XGBoost	0.183	0.629	0.888	0.012	0.997	0.09	0.586	0.635	0.580
DNN	0.230	0.695	0.890	0	1	0.12	0.680	0.612	0.688
Transformer	0.219	0.683	0.886	0.012	0.994	0.10	0.632	0.635	0.631

[§]Clinical characteristics include respiratory status, feeding status, and weight gain

[*]Cutoff refers to the predicted probability of TR-ROP from the specific ML model. When the predicted probability is higher than the cutoff, the infant was deemed to develop TR-ROP.

Table 7. Sensitivity, Specificity, and Area Under Receiver Operating Characteristic Curve From ML Models Only Using Demographics and Clinical Characteristics[§]

			Default Cutoff[*] (0.5)			Optimal Cutoff[*] (Highest Sen + Sp)			
ML Model	AUPRC	AUC	Accuracy	Se	Sp	Cutoff	Accuracy	Se	Sp
KNN	0.266	0.731	0.881	0.071	0.981	0.14	0.773	0.635	0.790
SVM	0.236	0.676	0.890	0.024	0.997	0.11	0.729	0.506	0.757
Random Forest	0.341	0.798	0.892	0.071	0.994	0.13	0.742	0.741	0.742
XGBoost	0.314	0.773	0.890	0.035	0.996	0.08	0.671	0.765	0.659
DNN	0.339	0.779	0.890	0.012	0.999	0.15	0.790	0.659	0.806
Transformer	0.275	0.745	0.886	0.059	0.988	0.11	0.625	0.765	0.608

[§]Demographics include BW, gestational age, gender, race and ethnicity; clinical characteristics include respiratory status, feeding status, and weight gain.

[*]Cutoff refers to the predicted probability of TR-ROP from the specific ML model. When the predicted probability is higher than the cutoff, the infant was deemed to develop TR-ROP.

However, the TR-ROP can still be missed clinically because some infants may miss crucial ROP examinations owing to being too ill, being transferred to another hospital or unit, or lack of an ophthalmologist trained to do ROP examinations. To decrease the number of eye examinations without missing cases of TR-ROP, the Imaging and Informatics in Retinopathy of Prematurity Consortium group developed and validated a risk prediction model that combined artificial intelligence with clinical demographics for predicting the development of TR-ROP.[16,17] The model using gestational age and vascular severity score had the highest performance with an AUC (AUPRC) of 0.82 (0.35) in the training dataset, and had an AUC of 0.92 in the external validation.[16,17] Different from their prediction model, we evaluated ML models with consideration of other factors, including feeding status, respiratory support, and the weight gain rate at 32 to 34 weeks PMA. We found these predictors improved the prediction for TR-ROP with an AUC of 0.853 and an AUPRC of 0.397. These studies demonstrate the potential use of ML models for early prediction of TR-ROP, which can facilitate the earlier identification of high-risk infants for close monitoring, timely intervention, or treatment of TR-ROP, thus improving the ROP outcomes. Additionally, it assists in resource allocation, ensuring that infants who require close monitoring or treatment for TR-ROP receive it promptly while minimizing the use of unnecessary resources on infants unlikely to develop TR-ROP.

Our previous investigation using a traditional multivariable logistic regression model showed that image evaluation findings of the posterior pole, ROP stage, zone, and retinal hemorrhage at the first image session at 32 to 34 weeks PMA predicted the development of TR-ROP with an AUC of 0.82, and the inclusion of demographics and clinical characteristics into prediction model improved AUC to 0.88.[12] The ML models did not improve AUCs than the traditional multivari-

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Table 8. The Performance of DNN for Predicting TR-ROP Using Various Predictors

Predictors	AUPRC	AUC	Optimal Cutoff (Highest Sen + Sp) Cutoff	Accuracy	Se	Sp
BW	0.293	0.762	0.12	0.646	0.812	0.625
GA	0.285	0.770	0.11	0.695	0.765	0.687
BW + GA	0.332	0.782	0.14	0.730	0.718	0.732
No. of quadrants with preplus	0.290	0.720	0.07	0.794	0.565	0.822
ROP stage + ROP zone	0.333	0.766	0.13	0.756	0.706	0.762
No. of quadrants with preplus + ROP stage + ROP zone	0.328	0.769	0.13	0.790	0.588	0.815
No. of quadrants with preplus + ROP stage + ROP zone + blot hemorrhage + pretinal hemorrhage	0.306	0.761	0.09	0.712	0.706	0.713
Respiratory support + feeding status	0.213	0.695	0.11	0.674	0.635	0.679
Weight growth rate	0.148	0.601	0.12	0.617	0.565	0.624
Respiratory support + feeding status + weight growth rate	0.234	0.686	0.12	0.691	0.612	0.701
GA + No. of quadrants with preplus	0.409	0.803	0.10	0.654	0.871	0.627
BW + GA + Number of quadrants with preplus	0.337	0.804	0.07	0.651	0.882	0.622
BW + GA + No. of quadrants with preplus + ROP zone + ROP stage	0.344	0.829	0.06	0.620	0.941	0.580
BW + GA + No. of quadrants with preplus + ROP zone + ROP stage + Blot hemorrhage + pretinal hemorrhage	0.407	0.834	0.12	0.799	0.741	0.806
BW + GA + No. of quadrants with preplus + ROP zone + ROP stage + blot hemorrhage + pretinal hemorrhage + weight gain rate	0.400	0.838	0.08	0.706	0.859	0.687
BW + GA + respiratory status+ feeding status + weight gain	0.342	0.795	0.12	0.735	0.729	0.736
BW + GA + respiratory status + feeding status + weight gain + image evaluation	0.372	0.828	0.18	0.803	0.729	0.812

BW, birth weight; GA, gestational age; ROP, retinopathy of prematurity.

able logistic regression model with AUCs up to 0.787 (DNN) using image evaluation findings, and up to 0.853 when including demographic and clinical characteristics. We found that image evaluation findings on the number of quadrants with preplus, ROP stage and zone, presence of blot hemorrhage and preretinal hemorrhage each contributed to the prediction of TR-ROP. We anticipate that deep learning of retinal images may lead to better prediction performance than traditional logistic regression models or ML models using image evaluation findings because deep learning of retinal images can better recognize various intricate patterns of vessel features and hemorrhages than human graders.[18]

Our study's methodology and findings are subject to several limitations that must be acknowledged. The primary constraint is the sample size and the cases of TR-ROP. Despite collecting data from 13 clinical centers in North America, the dataset is still relatively small for ML applications where larger datasets typically lead to improved accuracy and generalizability. The variations in image quality and characteristics of premature infants among the participating centers introduce heterogeneity that may impact the performance of ML models. Although hyperparameter optimization has been conducted, the extensive range of possible model configurations means that there may be unexplored combinations that could result in enhanced performance. The models used the trained reader's gradings of RetCam images (taken at 32–34 weeks PMA) instead of the findings from routine ROP examinations, which may limit the application of these models in clinical setting. However, with future improvement of these ML models using findings from routine ROP examinations or directly using retinal images, machine models may be applicable to the routine clinical care of ROP. This study is also limited in that validation in an external dataset was

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not performed owing to the lack of such data appropriate for external validation. To extend the applicability and robustness of our findings, future large multicenter and international studies are needed for broader validation of the ML models, ensuring the generalizability of models across varied clinical settings and populations.

In conclusion, this study extensively evaluated six commonly used ML models for early prediction of TR-ROP at a single early time point (32–34 weeks PMA) and demonstrated the modest performance of some ML models for potential clinical applications. Furthermore, we demonstrated the importance of adjusting the threshold of the predicted probability which can significantly impact the balance between sensitivity and specificity, depending on the clinical context. These findings underscore the importance of choosing the appropriate ML model based on the specific requirements of sensitivity and specificity, as well as the nature of the feature set used for the prediction.

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